causalgeo: channel effects from concurrent geo holdouts

Aditya Puttaparthi Tirumala[a,*]

[a]*Independent Researcher*

Abstract

Marketing teams sometimes turn off different bundles of media channels in different groups of markets at the same time. These geo holdouts measure each bundle's incremental effect, but not how that effect should be divided among its channels. Existing geo tools stop at the bundle. Marketing-mix models can split it, but the answer is unstable when channels move together.

`causalgeo` is a Python package for this problem. It assigns markets to concurrent cells, estimates bundle effects from randomized holdouts, and checks whether the design and impression data can support a channel-level split. It then fits channel slopes that are pulled toward the measured bundle totals, reports any gap between the model and experiment, and withholds slopes that the data cannot separate. In simulated outcomes built from known slopes, the linear recovery is within 0.21 percent. The package also shows when omitted carryover or saturation makes that simple model fail.

Keywords: geo experiment, incrementality, holdout, marketing mix, Python

1. Motivation and significance

Geo experiments estimate incremental impact by comparing markets assigned to a media change with similar control markets [1, 2]. This avoids the bias in comparing people who saw an ad with people who did not [3, 4]. It answers a simple question: did the channel create incremental sales? [5].

The number of markets that can go dark is usually limited. Consider a United States campaign across 210 designated market areas (DMAs). The

*Corresponding author.

Email address: `puttaparthy.aditya@gmail.com` (Aditya Puttaparthi Tirumala)

Table 1: Code metadata.

Nr.	Code metadata description	Value
C1	Current code version	0.1.0
C2	Permanent link to code/repository used for this code version	`https://github.com/adityapt/causalgeo/releases/tag/v0.1.0`
C3	Permanent link to Reproducible Capsule	`https://github.com/adityapt/causalgeo/blob/v0.1.0/paper/reproduce.py`
C4	Legal Code License	MIT
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python
C7	Compilation requirements, operating environments and dependencies	Python $\geq$ 3.10. Runtime dependencies: pandas, numpy, scipy, statsmodels. The example also uses matplotlib.
C8	Link to developer documentation	`https://github.com/adityapt/causalgeo/blob/main/README.md`
C9	Support email for questions	puttaparthy.aditya@gmail.com

business may allow a pause in only 30: large markets may be too risky, some regions may have partner commitments, and others may already be in a campaign. Those business rules set the holdout pool before the analyst can consider balance or power. Advertising effects are small compared with sales noise [6], so one well-powered single-channel test can use the whole pool. Teams therefore test bundles in several cells at once. Each cell identifies a bundle effect, not the effects of the channels inside it.

GeoLift and Meridian GeoX help design and measure single-cell tests [7, 8, 9, 10, 11]. They do not treat cells as channel bundles or connect a channel model back to the measured bundle total. Marketing-mix models use impression histories to estimate channel slopes, but collinear media makes the split arbitrary [12, 13]. Bayesian MMMs can use experimental results to set priors [14]; that can stabilize a split, but it cannot create information missing from the experiment or the exposure data.

`causalgeo` bridges the experiment and the channel model. It measures the bundle effect from the holdout, then uses that measurement to guide a linear model of channel impressions. The resulting channel slopes use both the experiment and the model; they are not direct experimental estimates. If the data cannot separate channels, the package reports the bundle instead. It does not control for test-period spend because the holdout itself changes spend [15].

2. Software description

2.1. Software architecture

The concurrent workflow, exposed through `design_concurrent_test` and `disentangle`, has four parts.

1. `design.py` assigns markets within strata, builds the cell-by-channel schedule, and reports which channels are separable.
2. `identified.py` estimates one bundle effect per cell and reports a direct channel effect only when that channel was tested alone.
3. `exposure.py` checks whether channels that should stay on changed differently in test and control markets.
4. `calibrated.py` fits the impression model and uses each measured bundle effect to guide the channel split.

Bundle effects use market and week fixed effects with market-clustered standard errors [16, 17, 18]. The supporting `analyze` function also provides time-based regression, synthetic control, and synthetic difference-in-differences for a single-cell test that has already been assigned. GeoLift and GeoX handle market selection and power search [7, 9].

2.2. Software functionalities

Design. The user supplies markets, strata, channels, and the channels to turn off in each cell. Within each stratum, `control_share` remains on; the other markets are divided among cells. The schedule has $A_{kc} = 1$ when cell k turns channel c off. The separation report calls a channel not tested, tied, separate, or not separable.

Bundle effect. Let $D_{it}^{(k)} = 1$ when market i is assigned to cell k and week t is in the test window. The cell regression is

$$Y_{it} = \sum_k \tau_k \, D_{it}^{(k)} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

τ_k is the cell's sales change after allowing for each market's history and common week shocks. It is a channel effect only when one channel was tested alone. Even if overlapping bundles cover all channels, a bundle coefficient does not belong to any one channel.

Still-on channels. A channel left on in cell k can still differ between the dark markets and the control. The impression shift is the difference in differences

$$S_{kc} = \big(\bar{X}_{c,k,\text{post}} - \bar{X}_{c,k,\text{pre}}\big) - \big(\bar{X}_{c,\text{control,post}} - \bar{X}_{c,\text{control,pre}}\big). \tag{2}$$

A pre-test slope b_c converts this impression shift into sales units. The usual cell estimate stays τ_k. If $|b_c S_{kc}|$ is more than 10 percent of the larger of $|\tau_k|$ and the pre-test outcome standard deviation, the channel split uses a refit that includes these still-on impressions. The user can also choose to report that refit. Turned-off channels are never controls: being off is the treatment. The example table reports the requested refit.

Channel split. Let W_{kc} be the impressions of channel c withheld in cell k,

$$W_{kc} = \max(-S_{kc},\, 0) \tag{3}$$

for channels that cell k turned off; all other entries in that calibration row are zero. The model should make $\sum_c \beta_c W_{kc}$ close to $-\tau_k$. This is a weighted observation, not a hard constraint. The impression model uses all pre-test observations and only control markets during the test; dark-market weeks are excluded. Precise bundle estimates receive more weight. The bundle table shows any remaining gap.

The reported covariance is the inverse weighted cross-product. It treats bundle estimates as independent, so it does not fully account for shared controls or reused observations. The simulations assess this approximation for this example design, not every possible panel.

Columns with cosine similarity above $1 - 10^{-6}$ are treated as copies. One stays in the model to account for their shared signal, but neither gets a slope. A slope is marked unstable if it changes substantially when the calibration

weight is reduced or when the market with the largest test-period outcome is removed. Missing impressions are not treated as zero. Daily impact is exposure times $\hat{\beta}_c$. The package offers geometric adstock but not saturation because one on/off holdout cannot trace a response curve [19, 13].

The output includes channel slopes, measured and model-implied bundle effects, the exposure check, and daily impacts. A market-cluster bootstrap reruns the pipeline on markets resampled within cells.

3. Illustrative examples

No public geo holdout ships with known channel-level effects. The example uses the published Meridian geo impression file `geo_all_channels.csv`: 40 markets and 156 weeks, from 25 January 2021 through 15 January 2024 [20]. Sales are constructed so the slopes are known. Meridian's conversions column is used only to form strata. It is not the outcome, because that series already contains Meridian's own media response.

Three of the five channels are used, with planted slopes 2×10^{-4}, 5×10^{-4}, and 1×10^{-4} sales per impression. Strata are quartiles of each market's mean conversions. With `control_share = 0.4` and seed 1, the design places 16 markets in control and 8 in each cell. The test window is the last 16 weeks, beginning 2 October 2023. In a dark market-week the turned-off channel's impressions are set to zero, and sales are

$$Y_{it} = \alpha_i + \gamma_t + \sum_c \beta_c X_{itc} + \varepsilon_{it}, \tag{4}$$

with $\alpha_i \sim N(0, 50^2)$, $\gamma_t \sim N(0, 20^2)$, and $\varepsilon_{it} \sim N(0, 30^2)$, drawn with seed 7. The script is `paper/example.py`.

The cells turn off {Channel0, Channel1}, {Channel0, Channel2}, and {Channel1, Channel2}, yielding rows $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 1)$. The schedule has full column rank, but no channel is tested alone; the assignment-only table therefore leaves all three channel effects blank. Table 3 instead reports the impression-model slopes, softly calibrated to the three measured bundles.

This is a recovery check for the linear model that the package fits, not evidence about saturation, long carryover, or the power of an eight-market cell. The impression paths are external to the package; sales are constructed from known slopes. Real experiments with smaller effects will have wider intervals [6].

Table 2: Cell effects on the constructed series, with the still-on refit requested. The default leaves the unadjusted column in place and only records the shift. The model column values the off-channel impression change at the fitted slopes.

Cell	Bundle	Unadjusted	Adjusted	SE	Model
1	Channel0+Channel1	−555.7	−587.2	71.4	−570.2
2	Channel0+Channel2	−318.0	−251.5	36.2	−253.6
3	Channel1+Channel2	−399.9	−399.9	54.4	−396.2

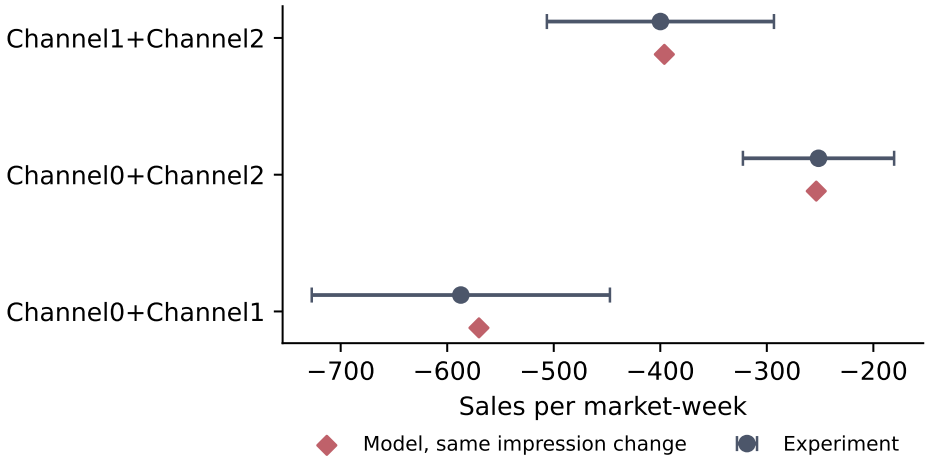


Figure 1: Experimental bundle effect, with a 95 percent interval, and the same off-channel impression change valued at the fitted slopes. Sales per market-week.

Cell 2 illustrates the exposure correction: Channel1's still-on shift is worth −83 sales, or 26 percent of the unadjusted effect, and refitting moves the estimate from −318.0 to −251.5. The corresponding shares are 13 percent for Cell 1 and 0.5 percent for Cell 3.

Table 2 compares experimental and model-implied bundle effects. Their gaps are 2.9, 0.8, and 0.9 percent of the experimental effects; all model totals lie within the experimental intervals. The bundle coefficient remains the experimental result, while the channel slope is the model-assisted split.

When Channel1 is replaced with a copy of Channel0, both are withheld. Their shared path remains as one control: Channel2 is then recovered as 0.97 sales per 10,000 impressions, versus its planted 1.00. Omitting the shared path instead biases it to 3.02. Figure 2 shows those two fits against the

Table 3: Planted and recovered slopes, in sales per 10,000 impressions. The largest absolute error is 0.21 percent.

Channel	Planted	Recovered	SE	Abs. error (%)
Channel0	2.000	1.997	0.006	0.17
Channel1	5.000	5.009	0.008	0.17
Channel2	1.000	1.002	0.008	0.21

Table 4: Two hundred refits per setting. Rho is the weight on Channel0 in the untreated Channel1 path. Error is the median absolute percent error among reported slopes. Coverage is the mean within-run coverage; brackets give its 95 percent Monte Carlo interval. At rho 1, error and coverage refer to the remaining channel.

Scale	Rho	Runs	Median error (%)	Coverage [MC interval]	Pair withheld
1.00	0.0	200	0.20	0.958 [0.942, 0.974]	0.00
1.00	0.9	200	0.40	0.953 [0.934, 0.973]	0.00
1.00	1.0	200	0.58	0.940 [0.907, 0.973]	1.00
0.25	0.0	200	0.77	0.958 [0.942, 0.974]	0.00
0.25	0.9	200	1.63	0.957 [0.938, 0.975]	0.00
0.25	1.0	200	2.23	0.940 [0.907, 0.973]	1.00

planted value.

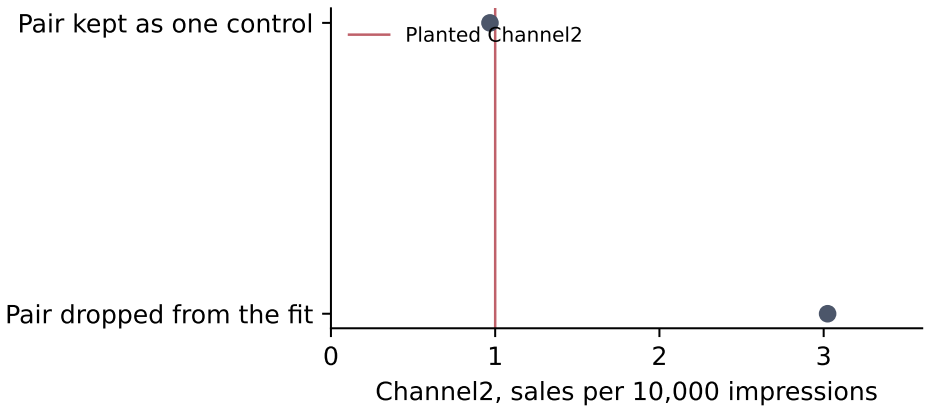


Figure 2: Channel2 when Channel1 is an exact copy of Channel0. Keeping the shared path as one control recovers the planted slope. Dropping that path pushes the remaining slope from about 1 to about 3 sales per 10,000 impressions.

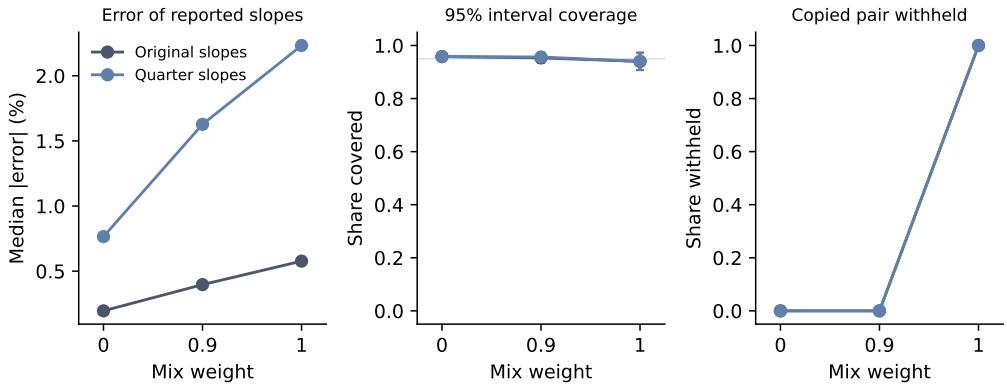


Figure 3: Recovery over 200 simulations per setting as the untreated Channel1 path is mixed toward Channel0. Bars in the coverage panel are 95 percent Monte Carlo intervals. At mix weight 1 the pair is withheld in every run, so error and coverage refer to the remaining channel.

Table 4 and Figure 3 summarize 200 noise draws for each slope scale and degree of Channel0–Channel1 similarity. At the original scale, median absolute error is 0.20 percent for distinct paths and 0.40 percent at a 0.9 mix; coverage is 95.8 and 95.3 percent. At an exact copy, the pair is withheld in every run and the remaining channel covers in 94.0 percent of runs (95 percent Monte Carlo interval: 90.7–97.3 percent). These results assess this design and the diagonal covariance approximation; they are not a general guarantee.

Table 5: Misspecification stress test over 200 draws. Error is the median absolute percent error in channel contribution to the experimental contrast. Coverage is the mean coverage of its nominal 95 percent interval; brackets give the 95 percent Monte Carlo interval. Gap is the median absolute model–experiment bundle gap as a percentage of the experimental effect.

Fit	Runs	Error (%)	Coverage [MC interval]	Gap (%)
Carryover omitted	200	47.26	0.000 [0.000, 0.000]	48.09
Carryover modeled	200	0.19	0.943 [0.924, 0.962]	1.03
Saturation omitted	200	27.00	0.000 [0.000, 0.000]	28.92

The preceding recovery check uses the model that generated sales. Table 5 instead evaluates two omitted response mechanisms over 200 draws. Geomet-

ric carryover has decay 0.5. The saturation response is $\beta_c X/(1+X/h_c)$, where h_c is twice the median positive exposure and β_c is the slope at zero. Because neither process has one true raw-impression slope, error and coverage refer to each channel's known contribution to the same pre/post, test/control contrast as the experiment. The carryover model is also refit with the planted decay to check the package's optional adstock transformation.

Supplying the planted carryover decay restores the recovery seen under the linear model. Omitting carryover or saturation produces large errors and no interval coverage. The corresponding bundle gaps are also large, so the package exposes the disagreement rather than making the misspecified channel split look experimentally validated. These stress tests do not select a decay or estimate a saturation curve.

Table 6: Further checks on the same design, 30 draws each. Serial correlation adds an AR(1) residual with autocorrelation 0.8. Heterogeneous slopes vary by market between half and one and a half times the planted slope. Redistribution moves half of a typical withheld week onto a channel the cell left on. Incomplete delivery leaves 30 percent of the control level running. The last two rows drop or keep the calibration rows on the original linear series.

Setting	Runs	Error (%)	Coverage	Gap (%)
Serial correlation	30	0.43	0.956	3.09
Heterogeneous slopes	30	4.80	0.356	4.27
Budget redistributed	30	0.22	0.967	12.60
Redistribution, refit requested	30	0.22	0.967	12.60
Incomplete delivery	30	0.22	0.967	3.84
Observational fit only	30	0.22	0.967	2.12
Calibrated fit	30	0.22	0.967	2.13

Table 6 asks what the linear recovery depends on. Serial correlation and a recorded shortfall in delivery leave the channel error near the linear result. Market-specific slopes do not: coverage falls to 35.6 percent. Moving the lost impressions onto a still-on channel leaves the channel error near 0.2 percent. The raw cell total sits 52.9 percent away from the total after those extra impressions are removed, because the experiment contains the reallocated delivery. The channel split is calibrated to the cleaned total, and the remaining bundle gap is 12.6 percent. Replacing the reported cell estimate with that refit does not change the split, so the two budget rows agree. On the original series the observational fit and the calibrated fit agree,

so those slopes are coming from the impression histories. The calibration rows are not what produces the 0.2 percent recovery. Figure 4 puts that pattern next to the omitted-response checks.

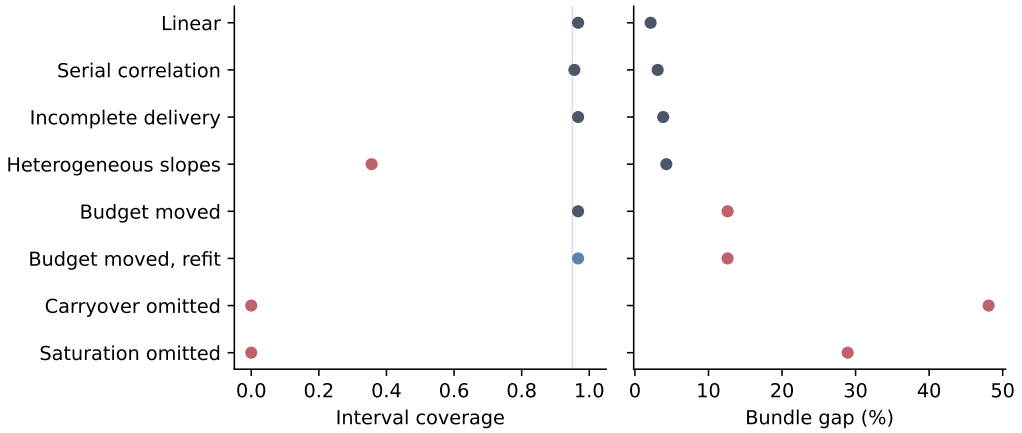


Figure 4: Coverage and the bundle gap. The linear, serial, delivery, slope, and budget rows are 30 draws. Carryover and saturation are the 200-draw checks in Table 5. The pale line is 95 percent coverage.

4. Impact

The package gives teams one place to assign cells, measure bundle effects, check which channels can be separated, and report model-assisted channel slopes next to the remaining model–experiment gap. Existing open geo tools stop at the cell [7, 9]. Analysts needing a channel number otherwise split the result in a separate mix model [12, 13], often without reporting its gap from the experiment or refusing to split channels that moved together. Here the bundle is the directly measured result; the channel is the finer, model-assisted result.

The current release is 0.1.0. Adoption is early, and this paper has no user study. The reusable example, `paper/reproduce.py`, rebuilds all five tables and four figures. The same functions apply to a real holdout once the user supplies market outcomes and channel impressions. Channels the design cannot separate, or whose impressions are copies, remain blank rather than receiving a made-up slope.

5. Conclusions

`causalgeo` measures concurrent channel-bundle holdouts and splits a measured bundle only when the impression data support it. The calibration is soft: it guides the channel model toward the experiment rather than forcing agreement. On published Meridian paths with constructed outcomes, the three slopes are recovered within 0.21 percent and their bundle totals are 0.8 to 2.9 percent from the experimental effects. A copied pair is kept as one control path and is not split. Across 200 simulations per setting, coverage ranges from 94.0 to 95.8 percent. A matched carryover fit gives 94.3 percent coverage; omitted carryover or saturation produces large channel errors and bundle gaps. Intervals remain conditional on the response model and the diagonal covariance approximation.

The split is linear, with optional geometric adstock and no saturation curve [13]. A channel being on or off does not reveal the full response curve, so the package does not estimate one. GeoLift and GeoX handle market selection and power [7, 9]; this package assigns markets within strata the user has already chosen, and does not claim that eight markets make a powered cell [6]. The default cell result is the unadjusted experiment. When lost impressions move to a channel left on, the split uses the refitted cell total after that delivery is removed. Test-period spend stays out of the regression because assignment changes it [15, 3].

CRediT authorship contribution statement

Aditya Puttaparthi Tirumala: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The impression paths are the Meridian file `geo_all_channels.csv` [20]. In `paper/`, set `PYTHONPATH=../src` and run `python3.11 reproduce.py`. That rebuilds Tables 2–6 and the four figures, then checks them against the rounded values reported here. Table 1 gives the repository links.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Cursor, with the Grok language model, to draft and revise the manuscript and the example code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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